

Python Automation: Essential Reference

Technical Documentation

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File Operations (I/O)

Reading a File

Python can read files using the open function with the "r" mode.

```
with open("file.txt", "r") as f:
    data = f.read()
```

Writing to a File

Python can write to files using the open function in write ("w") mode. This overwrites existing content.

```
with open("file.txt", "w") as f:
    f.write("Hello world")
```

Appending to a File

Adds new content to the end of an existing file without deleting current data.

```
with open("file.txt", "a") as f:
    f.write("\nNew line")
```

Reading Lines

Iterates through a file line by line, which is memory-efficient for large files.

```
with open("file.txt", "r") as f:
    for line in f:
        print(line)
```

System & Process Automation

Running Commands via Python

The subprocess module allows Python to execute terminal commands and capture output.

```
import subprocess
subprocess.run(["git", "status"])
```

Executing Python Script via Subprocess

Run an external Python script from within another Python script.

```
import subprocess
subprocess.run(["python3", "script.py"])
```

File System Management (os module)

Listing Files in Python

Retrieve a list of all files and directories in a specified path.

```
import os
files = os.listdir(".")
print(files)
```

Checking File Existence

Verify if a specific file exists on the disk before performing operations.

```
import os
exists = os.path.isfile("file.txt")
```

Creating Directory

Create a new folder at the specified path.

```
import os
os.mkdir("my_folder")
```

Renaming or Moving a File

Change the name of a file or move it to a different directory.

```
import os
os.rename("old.txt", "new.txt")
```

Removing a File

Permanently delete a file from the file system.

```
import os
os.remove("file.txt")
```

Error Handling

Handling Exceptions

Safely handle runtime errors to prevent the script from crashing.

```
try:
    with open("file.txt", "r") as f:
        content = f.read()
except FileNotFoundError:
    print("Error: The file was not found.")
```